Physics 1 – Lecture Notes

Kinematics

Alessandro Candeloro

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1 Kinematics: the study of motion

Kinematics (from the Greek *kinema*, "movement") is the branch of mechanics that describes the *motion* of bodies, without asking what causes it. It answers questions like "where will the ball be in two seconds?" or "how long does the car need to stop?". Questions like "why does the ball fall?" belong to *dynamics*, which we will meet later. Kinematics comes first because we cannot formulate the laws of motion until we have a precise language for talking about motion.

That language is mathematics. To give a *mathematical description* of motion we have to turn "where the body is" into numbers, and for that we need a **coordinate system**. A coordinate system, in turn, only makes sense once we have said *relative to what* we are measuring: it is always attached to a **reference frame**.

1.1 Reference frames

Is a passenger sitting in a moving train at rest or in motion? Both answers are correct, depending on who is looking. For the passenger next to her she is at rest; for someone standing on the platform she is moving at the speed of the train. Motion is *relative*: it only has a meaning with respect to something else.

A reference frame makes this "something else" explicit. It is *observer-centric*: it is attached to whoever is observing the motion. For example:

- if I am sitting in a car, the natural reference frame is fixed with the car. In this frame the dashboard is at rest, while the trees outside move backwards;
- for a person standing at a traffic light, the reference frame is fixed with the terrain. In this frame the trees are at rest and the car moves forward.

Neither description is "more true" than the other. They describe the same physical situation from two different points of view, and the numbers we find (positions, velocities) differ between the two frames.

Once the frame is chosen, the *coordinate system* is the set of mathematical labels we put on it: where we place the origin, which direction we call positive, and which unit of length we use. Different coordinate systems in the same frame describe the same motion with different numbers, but these numbers are related in a simple way (for example, by shifting the origin).

Rule

Every time we describe a physical motion we must first fix **a reference frame** *and* **a coordinate system**. Without them, statements such as "the body is at $x = 3\,\mathrm{m}$" or "it moves at $5\,\mathrm{m/s}$" have no meaning.

1.2 The point-particle model

Real objects are complicated: a car has wheels that rotate, a thrown ball spins, a person moves arms and legs while walking. Describing all of this at once would be hopeless as a first step. So we make a *model*: complex (extended) objects will be treated as *points*, as if the whole object were concentrated in one place.

This is an approximation, and it works well whenever the size of the object is small compared to the distances it travels, and we are not interested in its rotation or deformation. To describe a

car driving from Palermo to Catania, the length of the car (a few metres) is irrelevant compared to the roughly 200 km of the trip. Even the Earth can be treated as a point when we study its orbit around the Sun.

Once a body is a point, its motion is fully described by (at least) three physical variables, all functions of time:

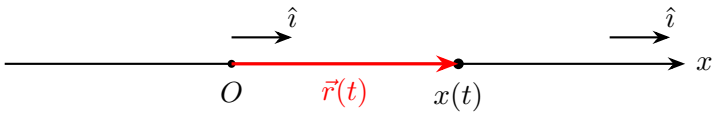
$$\vec{x}(t) \text{ (position)}, \qquad \vec{v}(t) \text{ (velocity)}, \qquad \vec{a}(t) \text{ (acceleration)}.$$

The rest of this lecture is about defining these three quantities carefully and understanding how they are related.

2 One-dimensional kinematics

We start with the simplest possible case: motion along a straight line, such as a train on a straight track, a car on a straight road, or a stone dropped vertically. This is *one-dimensional* motion. It is simple because a single number is enough to say where the body is, but it already contains all the important ideas, and in the next lectures we will see that motion in two and three dimensions can be built from it.

We choose an axis x along the line of motion, with origin O and unit vector $\hat{\imath}$ pointing in the positive direction. The unit vector $\hat{\imath}$ has length 1 and no units: its only job is to indicate a direction.



In one dimension every vector is a number times $\hat{\imath}$, and the *sign* of that number carries the information about direction: positive means "along $\hat{\imath}$", negative means "opposite to $\hat{\imath}$".

2.1 Position

The **position coordinate** is defined with respect to the chosen reference frame / coordinate system. At a given time t the position vector is the arrow from the origin to the body:

$$\vec{r}(t) = x(t)\,\hat{\imath}.$$

The number $x(t)$ is the *coordinate* of the body: it tells us how far the body is from the origin, and on which side. A body at $x = -2\,\text{m}$ is two metres from O on the negative side.

We usually call the instant at which we start observing $t = 0$. The position at that moment is called the *initial position*:

$$\vec{r}(0) = x(t=0)\,\hat{\imath} = x_0\,\hat{\imath},$$

Position is a length $[\vec{r}] = L$ and it si measured in the SI system in metres (m).

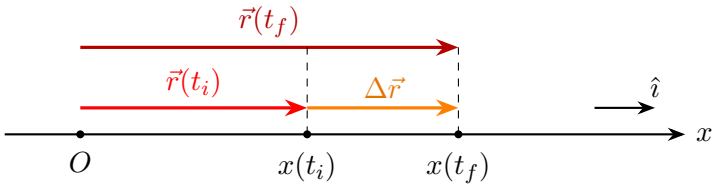
Note that the value of x depends on where we put the origin. Moving the origin changes every position coordinate by the same amount. This is a hint that position on its own is not the most "physical" quantity. What matters is how position *changes*.

2.2 Displacement

The **displacement** is the *change in position* of a body. Suppose we observe the body at position $x(t_i)$ at time $t = t_i$ ("initial") and at position $x(t_f)$ at a later time $t = t_f > t_i$ ("final"). The displacement is the arrow that goes from the initial position to the final one:

$$\begin{aligned}\Delta\vec{r} &= \vec{r}(t_f) - \vec{r}(t_i) = \\ &= x(t_f)\,\hat{\imath} - x(t_i)\,\hat{\imath} \\ &= \big(x(t_f) - x(t_i)\big)\hat{\imath} = \\ &= \Delta x\,\hat{\imath}.\end{aligned}$$

The symbol Δ ("delta") always means *final value minus initial value*, and here it follows a pictorial representation of the displacement vector in 1D.



Two things are worth noticing:

- The displacement does *not* depend on where we put the origin. If we shift the origin, both $x(t_i)$ and $x(t_f)$ change by the same amount, and their difference stays the same. This is why displacement is more physical than position.
- The displacement only depends on the start and end points, not on what happens in between. A body that goes forward 10 m and then back 4 m has the same displacement, $\Delta x = +6$ m, as one that simply moves forward 6 m.

The second point will be important when we compare velocity and speed.

3 Velocity and speed

Knowing that a body has moved by 6 m does not tell us much unless we also know how long it took: 6 m in one second is a sprint, 6 m in an hour is a snail. To capture "how fast" the position changes, we divide the displacement by the time interval.

3.1 Average velocity

The first concept we introduce is the average velocity. Let $\Delta t = t_f - t_i$ and $\Delta\vec{r} = \Delta x\,\hat{\imath}$.

Average velocity

The **average velocity** over the interval Δt is

$$\vec{v}_{\text{avg}} = \frac{\Delta \vec{r}}{\Delta t} = \frac{\Delta x}{\Delta t}\,\hat{\imath}\,.$$

The dimension of velocity is

$$[\vec{v}_{\text{avg}}] = \frac{L}{T}$$

and it is units of measure are meter per second m/s or kilometer per hour Qkm/h.

Since $\Delta t > 0$ (time goes forward), the average velocity is a vector pointing in the same direction as the displacement. Its component $v_{\text{avg},x} = \Delta x/\Delta t$ can therefore be either positive or negative:

$$v_{\text{avg},x} > 0 \iff \Delta x > 0 \Rightarrow \text{positive displacement},$$
$$v_{\text{avg},x} < 0 \iff \Delta x < 0 \Rightarrow \text{negative displacement}.$$

A negative velocity does not mean anything strange: it simply means the body is, on average, moving in the direction opposite to $\hat{\imath}$.

Worked example

A car on a straight road passes the kilometre marker $x(t_i) = 2\,\text{km}$ and, 10 minutes later, the marker $x(t_f) = 12\,\text{km}$. Its average velocity is

$$v_{\text{avg},x} = \frac{12\,\text{km} - 2\,\text{km}}{10\,\text{min}} = \frac{10\,\text{km}}{\frac{1}{6}\,\text{h}} = 60\,\text{km/h}.$$

To convert to SI units, recall that $1\,\text{m/s} = 3.6\,\text{km/h}$, so $60\,\text{km/h} \approx 16.7\,\text{m/s}$. If the car were driving the other way, from marker 12 to marker 2, we would find $v_{\text{avg},x} = -60\,\text{km/h}$: same rate, opposite direction.

The word "average" is important. The car may have stopped at a traffic light and then sped up to make up time; the average velocity hides all these details and only looks at where the car started and where it ended.

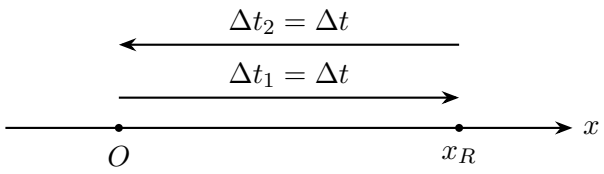
3.2 Speed

In everyday language "speed" and "velocity" mean the same thing. In physics they do not. The **speed** measures how much *distance* is covered per unit time, regardless of direction. If during the interval $\Delta t = t_f - t_i$ the body travels a total path length d (what a car's odometer would record), the (average) speed is

$$s = \frac{d}{\Delta t} \geq 0, \qquad d \geq 0.$$

The speed is *never negative*, and it is zero only if the body does not move at all. For motion in a single direction (no turning back), the path length equals the size of the displacement, $d = |\Delta \vec{r}| = |\Delta x|$.

Let us compare speed and average velocity to highlight why they are different: **speed** is a *scalar*. It counts every metre travelled, forward or backward; **average velocity** $\vec{v}_{\text{avg}} = \dfrac{\Delta \vec{r}}{\Delta t} = \dfrac{\Delta x}{\Delta t}\,\hat{\imath}$ is a *vector*. It only cares about the net change in position. The clearest way to see the difference is a round trip. A body goes from O to x_R in a time $\Delta t_1 = \Delta t$, and comes back from x_R to O in a time $\Delta t_2 = \Delta t$.



The body has travelled the distance x_R twice, but it has ended where it started. So the total path length and the total displacement are

$$d = 2x_R, \qquad \Delta\vec{r} = 0\,\hat{\imath},$$

and over the whole trip, which lasts $2\Delta t$,

$$s = \frac{2x_R}{2\Delta t} = \frac{x_R}{\Delta t} \neq 0, \qquad \vec{v}_{\text{avg}} = \frac{\Delta\vec{r}}{2\Delta t} = 0\,\hat{\imath} \quad \Longrightarrow \quad |\vec{v}_{\text{avg}}| \neq s\,!$$

Worked example

A swimmer swims one length of a 50 m pool in 25 s and comes back to the starting wall in another 25 s. Her average speed is $s = 100\,\text{m}/50\,\text{s} = 2\,\text{m/s}$, which is a good measure of how hard she was swimming. Her average velocity over the full 50 s is zero, because she is back where she started. Both numbers are correct; they just answer different questions.

Remark

Only if $|\Delta\vec{r}|$ coincides with the path length d (i.e. the body never reverses its direction) do we have $|\vec{v}_{\text{avg}}| = s$.

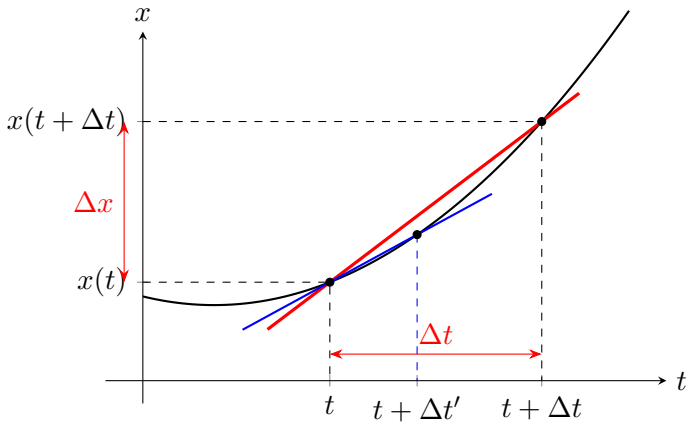
3.3 Instantaneous velocity

The average velocity describes a whole time interval with a single number. But when you look at a car's speedometer, you read the velocity *at that instant*, not an average over the whole trip. How can we define a velocity at a single instant, when velocity is "displacement divided by time" and in a single instant there is neither displacement nor elapsed time?

The answer is to take average velocities over shorter and shorter intervals. So far we considered a *discrete* time interval $[t_1, t_2]$. Take now the interval $[t, t+\Delta t]$, starting at the instant t we are interested in, with positions $x(t)$ and $x(t+\Delta t)$. The average velocity over this interval is

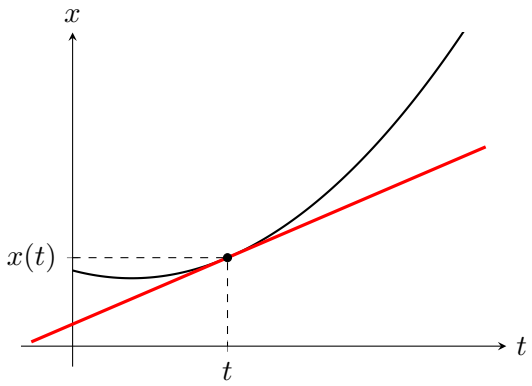
$$\vec{v}_{\text{avg}}(t) = v_{\text{avg},x}(t)\,\hat{\imath}, \qquad v_{\text{avg},x}(t) = \frac{\Delta x}{\Delta t} = \frac{x(t+\Delta t) - x(t)}{\Delta t}.$$

It helps to look at this on a graph of position against time. The quantity $\Delta x/\Delta t$ is "rise over run": it is the *slope of the secant line* through the points $(t, x(t))$ and $(t+\Delta t, x(t+\Delta t))$ on the graph of $x(t)$. If we take a smaller interval $\Delta t' < \Delta t$ (blue line), the secant rotates and gets closer to the curve near t.



If we keep shrinking the interval, $\Delta t \to 0$, the second point slides along the curve towards the first, and the secant becomes the *tangent* to the curve $x(t)$ at time t. The limit of the average velocity is what we call the instantaneous velocity:

$$\lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \lim_{\Delta t \to 0} \frac{x(t+\Delta t) - x(t)}{\Delta t} = v_x(t).$$



Instantaneous velocity

$$\vec{v}(t) = \lim_{\Delta t \to 0} \frac{\Delta \vec{r}}{\Delta t} = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t}\,\hat{\imath} = v_x(t)\,\hat{\imath}, \qquad v_x(t) = \frac{\mathrm{d}}{\mathrm{d}t}\,x(t).$$

The instantaneous velocity is the *derivative* of the position with respect to time, i.e. the *slope of the tangent* to $x(t)$ at time t.

Worked example

Suppose a body moves according to $x(t) = t^2$ (with x in metres and t in seconds), and we want its velocity at $t = 1\,\text{s}$. Let us compute the average velocity over shorter and shorter intervals starting at $t = 1\,\text{s}$:

Δt (s)	$x(1 + \Delta t)$ (m)	Δx (m)	$\Delta x/\Delta t$ (m/s)
1	4	3	3
0.1	1.21	0.21	2.1
0.01	1.0201	0.0201	2.01
0.001	1.002001	0.002001	2.001

The average velocities approach 2 m/s, and indeed $v_x(t) = \dfrac{\mathrm{d}}{\mathrm{d}t}\, t^2 = 2t$, which gives 2 m/s at $t = 1\,\text{s}$. The derivative is nothing mysterious: it is just the value the average velocity settles on as the interval becomes very short.

The sign of v_x tells us the direction of motion at that instant (for small Δt):

- $v_x(t) > 0 \Rightarrow x(t + \Delta t) > x(t)$: the coordinate is increasing, so we are moving *along* the direction of $\hat{\imath}$;
- $v_x(t) < 0 \Rightarrow x(t + \Delta t) < x(t)$: the coordinate is decreasing, so we are moving in the direction *opposite* to $\hat{\imath}$.

On the graph: where the curve $x(t)$ goes up, the velocity is positive; where it goes down, the velocity is negative; at a peak or a valley (where the tangent is horizontal) the velocity is zero, and the body is momentarily at rest, as when a ball thrown upwards reaches its highest point.

From now on, when we say "velocity" without further qualification we mean the instantaneous velocity. Its magnitude $|\vec{v}(t)|$ is the *instantaneous speed*, which is what the speedometer shows.

3.4 Particle under constant velocity

The simplest kind of motion is one where the velocity never changes: a car driving with cruise control on a straight, flat road. Constant velocity means the *vector* is constant: same magnitude *and* same direction. On an $x(t)$ graph this motion is a straight line, because the slope is the same everywhere.

If the particle moves with constant velocity $\vec{v}$, the velocity at every instant is the same, so it must also equal the average velocity over *any* interval:

$$\vec{v}(t) = \vec{v} = \vec{v}_{\text{avg}} = \frac{\Delta \vec{x}}{\Delta t} = \frac{\vec{x}(t_f) - \vec{x}(t_i)}{t_f - t_i}.$$

Multiplying both sides by $(t_f - t_i)$:

$$\vec{v}\,(t_f - t_i) = \vec{x}(t_f) - \vec{x}(t_i) \quad \Longrightarrow \quad \vec{x}(t_f) = \vec{x}(t_i) + \vec{v}\,(t_f - t_i).$$

This simple formula has an important consequence. If we fix the initial position $\vec{x}(t_i)$ and we know $\vec{v}$, we can *predict* the position at any time t_f, in the future or in the past:

$$x(t_f) = x(t_i) + v_x\,(t_f - t_i).$$

This is the **equation of motion for constant velocity**, and it is the first example of what physics is about: from a few pieces of information (where the body is now and how it moves) we can say where it will be. A standard notation is $t_i = 0$, $\vec{x}(t_i) = \vec{x}_0$ and $t_f = t$, with which we get

$$\vec{x}(t) = \vec{x}_0 + \vec{v}\,t$$

Worked example

A cyclist is at $x_0 = 100\,\text{m}$ at $t = 0$ and rides towards the origin with constant velocity $v_x = -5\,\text{m/s}$ (negative because she moves opposite to $\hat{\imath}$). Her position is $x(t) = 100\,\text{m} - (5\,\text{m/s})\,t$. At $t = 8\,\text{s}$ she is at $x = 60\,\text{m}$; she reaches the origin when $x(t) = 0$, i.e. at $t = 20\,\text{s}$.

4 Acceleration

In most real motions the velocity is not constant. A car pulls away from a traffic light, a ball slows down as it rises, a train brakes as it arrives at the station. Just as velocity measures how fast the *position* changes, we need a quantity that measures how fast the *velocity* changes. This is the acceleration. The construction follows exactly the same steps we used for velocity: first an average over an interval, then a limit.

4.1 Average acceleration

The average acceleration describes how the instantaneous velocity changes in a time interval $[t_1, t_2]$:

$$\vec{a}_{\text{avg}} = a_{\text{avg},x}\,\hat{\imath} = \frac{\Delta \vec{v}_x}{\Delta t} = \frac{\vec{v}(t_2) - \vec{v}(t_1)}{t_2 - t_1} = \frac{v_x(t_2) - v_x(t_1)}{t_2 - t_1}\,\hat{\imath}\,.$$

The corresponding dimensions are

$$[\vec{a}_{\text{avg}}] = \frac{[v_x]}{[\Delta t]} = \frac{L}{T}\cdot\frac{1}{T} = \frac{L}{T^2} \quad (\text{m/s}^2).$$

The unit m/s^2 is best read as "metres per second, *per second*": an acceleration of $3\,\text{m/s}^2$ means that the velocity changes by $3\,\text{m/s}$ every second.

Worked example

A car goes from rest to 100 km/h in 10 s. Converting, $100\,\text{km/h} = 100/3.6\,\text{m/s} \approx 27.8\,\text{m/s}$, so

$$a_{\text{avg},x} = \frac{27.8\,\text{m/s} - 0}{10\,\text{s}} \approx 2.8\,\text{m/s}^2.$$

For comparison, a body falling freely near the Earth's surface has an acceleration of about $9.81\,\text{m/s}^2$: its velocity grows by almost $10\,\text{m/s}$ every second, more than three times faster than the car.

4.2 Instantaneous acceleration

As with velocity, we obtain the acceleration at a single instant by shrinking the interval. Take $t_1 = t$ and $t_2 = t + \Delta t$:

$$\vec{a}_{\text{avg}}(t) = \frac{\vec{v}(t + \Delta t) - \vec{v}(t)}{\Delta t} \quad \xrightarrow{\Delta t \to 0} \quad \vec{a}(t).$$

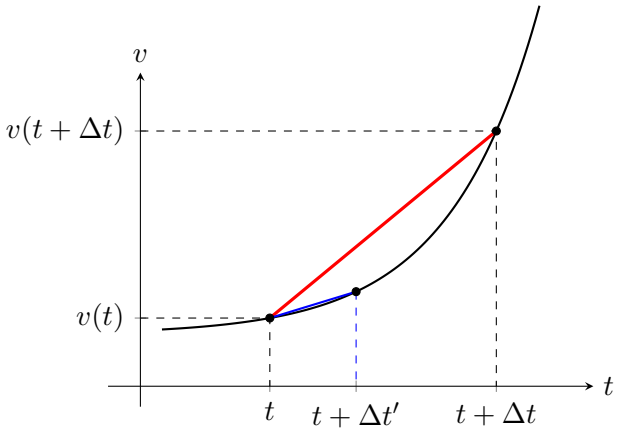
Instantaneous acceleration

$$\vec{a}(t) = \frac{\mathrm{d}}{\mathrm{d}t}\,\vec{v}(t) = \frac{\mathrm{d}^2}{\mathrm{d}t^2}\,\vec{x}(t).$$

The acceleration is the derivative of the velocity, and therefore the *second* derivative of the position. Note the chain:

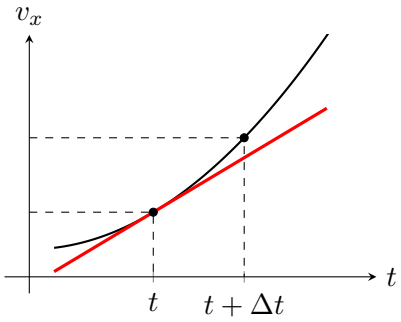
$$x(t) \xrightarrow{\frac{\mathrm{d}}{\mathrm{d}t}} v_x(t) \xrightarrow{\frac{\mathrm{d}}{\mathrm{d}t}} a_x(t).$$

The graphical picture is the same as before, with $v(t)$ in place of $x(t)$. As $\Delta t \to 0$, the average acceleration (slope of the secant on the $v(t)$ graph) approaches the instantaneous acceleration, which corresponds to the *slope of the tangent* to the curve $v(t)$ at time t.

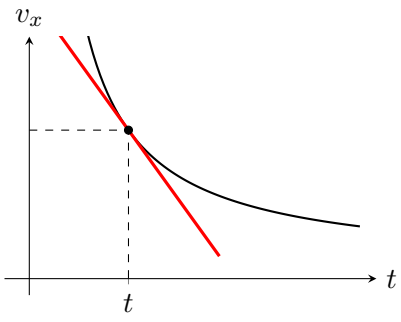


Sign of the acceleration

Just as the sign of v_x tells us whether x is increasing or decreasing, the sign of a_x tells us whether the *velocity component* v_x is increasing or decreasing.



$a_x(t) > 0 \;\Rightarrow\; v_x(t + \Delta t) > v_x(t)$ (for small Δt).
Positive acceleration: the velocity component in the direction $\hat{\imath}$ is *increasing*. The tangent to $v_x(t)$ goes up.



$a_x(t) < 0 \;\Rightarrow\; v_x(t + \Delta t) < v_x(t)$ (for small Δt).
Negative acceleration: the velocity component in the direction $\hat{\imath}$ is *decreasing*. The tangent to $v_x(t)$ goes down.

Here comes the most common source of confusion in this topic. In everyday language "deceleration" means "slowing down", and it is tempting to think that a negative acceleration always means slowing down.

Negative acceleration $\neq$ slowing down

Velocity and acceleration are *vector* quantities, so the sign of a_x alone does not tell whether the body is speeding up or slowing down. What matters is how $\vec{a}$ is oriented *relative to* $\vec{v}$:

- $\vec{a}$ and $\vec{v}$ point in the **same** direction $\Rightarrow$ the body is **speeding up**;
- $\vec{a}$ and $\vec{v}$ point in **opposite** directions $\Rightarrow$ the body is **slowing down**.

The reason is that "speeding up" and "slowing down" refer to the *speed* $|v_x|$, while the sign of a_x refers to the *component* v_x. If $v_x = -10\,\mathrm{m/s}$ becomes $v_x = -15\,\mathrm{m/s}$, the component has *decreased* ($a_x < 0$), but the speed has gone from 10 to 15 m/s: the body is moving faster. The four possible cases are:

v_x	a_x	$\vec{a}$ vs. $\vec{v}$	The body…
$+$	$+$	same direction	moves along $+\hat{\imath}$ and speeds up
$+$	$-$	opposite direction	moves along $+\hat{\imath}$ and slows down
$-$	$-$	same direction	moves along $-\hat{\imath}$ and speeds up
$-$	$+$	opposite direction	moves along $-\hat{\imath}$ and slows down

Worked example

A car drives in the $-\hat{\imath}$ direction with $v_x = -20\,\mathrm{m/s}$ and brakes with constant acceleration of magnitude $4\,\mathrm{m/s^2}$. Since it is slowing down, $\vec{a}$ must point opposite to $\vec{v}$, i.e. along $+\hat{\imath}$: $a_x = +4\,\mathrm{m/s^2}$. A *positive* acceleration, for a car that is braking. After 5 s its velocity is $-20 + 4 \cdot 5 = 0$: the car has stopped.

5 Uniform rectilinear motion

Let us go back to the particle moving with constant velocity and compute its acceleration. If $\vec{v}(t) = \vec{v}$ for every t, then $\vec{v}(t + \Delta t) = \vec{v}$ as well. Hence

$$\vec{v}(t + \Delta t) - \vec{v}(t) = 0 \quad \Longrightarrow \quad \vec{a}(t) = 0.$$

This is not surprising: acceleration measures change in velocity, and here the velocity does not change. But it is worth stating clearly, because the converse is also true and will be very important in dynamics: a body with zero acceleration either stays at rest or moves in a straight line at constant speed.

Uniform rectilinear motion

The acceleration of a particle under constant velocity is zero. This motion is called **uniform rectilinear motion**: *uniform* because the speed does not change, *rectilinear* because the path is a straight line.

6 Uniformly accelerated linear motion

The next step up in complexity is a motion in which the velocity changes, but in the simplest possible way: at a constant rate. Near the Earth's surface, every body falling freely (if we ignore air resistance) moves with constant acceleration, and so, to a good approximation, does a car accelerating steadily or braking.

Suppose that the acceleration is constant (as a *vector*). If the acceleration never changes, then its average over any interval is the same as its instantaneous value:

$$\vec{a}(t) = \text{const} = \vec{a} = \vec{a}_{\text{avg}}(t) = \frac{\Delta\vec{v}}{\Delta t}.$$

This is exactly the same trick we used for constant velocity, one level higher: there, $\vec{v} = \vec{v}_{\text{avg}}$ gave us the position; here, $\vec{a} = \vec{a}_{\text{avg}}$ will give us the velocity.

6.1 Velocity as a function of time

From the definition,

$$\vec{a} = \frac{\Delta\vec{v}}{\Delta t} = \frac{\vec{v}(t_f) - \vec{v}(t_i)}{t_f - t_i} \quad \Longrightarrow \quad \vec{a}\,(t_f - t_i) = \vec{v}(t_f) - \vec{v}(t_i),$$

that is

$$\vec{v}(t_f) = \vec{v}(t_i) + \vec{a}\,(t_f - t_i).$$

With $t_i = 0$, $\vec{v}(0) = \vec{v}_0$ (the *initial velocity*) and $t_f = t$:

$$\vec{v}(t) = \vec{v}_0 + \vec{a}\,t \quad \Longrightarrow \quad v_x(t)\,\hat{\imath} = v_{x,0}\,\hat{\imath} + a_x\,\hat{\imath}\,t,$$

and taking the $\hat{\imath}$ component,

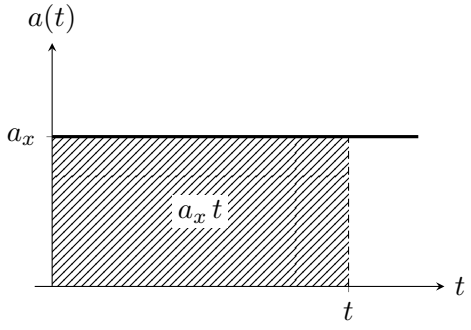
$$v_x(t) = v_{x,0} + a_x\,t$$

The velocity is a linear function of time: on a $v(t)$ graph it is a straight line with slope a_x, crossing the vertical axis at $v_{x,0}$.

There is a very useful way to read this result on a graph of acceleration against time. Since a_x is constant, the graph of $a(t)$ is a horizontal line, and the region under it between 0 and t is a rectangle of height a_x and width t. Its area is

$$\text{Area} = a_x\,t = v_x(t) - v_{x,0} = \Delta v.$$

The change in velocity equals the *area under the $a(t)$ curve.*

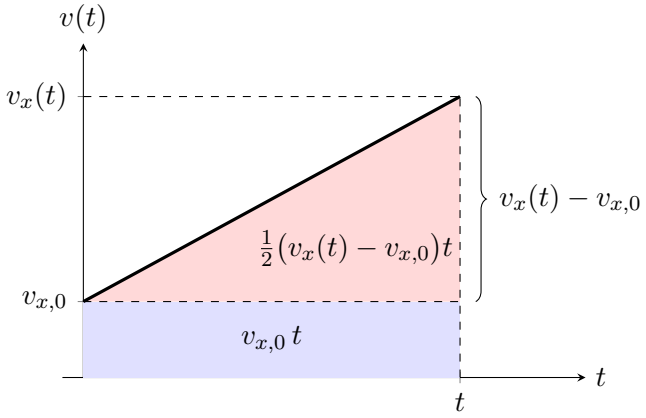


Intuition

Why should an area give a change? Split the interval $[0, t]$ into many small pieces of duration δt. In each piece the velocity changes by about $a\,\delta t$, which is the area of a thin vertical strip of height a and width δt. Adding up all the small changes gives the total change, and adding up all the strips gives the total area. This idea works even when the acceleration is *not* constant (the strips just have different heights), and it is the physical meaning of the integral: $\Delta v = \int a\,\mathrm{d}t$.

6.2 Position as a function of time

The same reasoning, one level down, gives us the position. Velocity is the rate of change of position, so the displacement equals the *area under the $v(t)$ curve*. Since $v_x(t) = v_{x,0} + a_x t$ is a straight line, the area between 0 and t is easy to compute: it is a rectangle plus a triangle.



The rectangle (blue) has height $v_{x,0}$ and width t: it is the distance the body would cover if it kept its initial velocity. The triangle (red) has base t and height $v_x(t) - v_{x,0}$: it is the *extra* distance gained because the body is accelerating. Adding them:

$$\begin{aligned}\text{Area} &= v_{x,0}\,t + \tfrac{1}{2}\big(v_x(t) - v_{x,0}\big)\,t \\ &= v_{x,0}\,t + \tfrac{1}{2}\,a_x\,t^2,\end{aligned}$$

where we used $v_x(t) - v_{x,0} = a_x t$.

We can reach the same result without areas. Since v_x grows linearly in time, it spends as much time above its midpoint value as below it, so the average velocity over $[0, t]$ is simply the mean of the initial and final values.

$$v_{\text{avg},x}(t) = \frac{v_{x,0} + v_x(t)}{2} = v_{x,0} + \frac{a_x t}{2} = \left.\frac{\Delta x}{\Delta t}\right|_{\Delta t = t}.$$

Average velocity = arithmetic mean: only for constant acceleration

The average velocity equals the arithmetic mean of the initial and final velocities *only* when the acceleration is constant, because only then is $v(t)$ a straight line. Suppose a car starts at 100 km/h and ends, one hour later, at 20 km/h. If it slows down steadily, its average velocity is indeed $(100 + 20)/2 = 60$ km/h. But if it brakes hard to 20 km/h in the first minute and then cruises at 20 km/h for the remaining 59 minutes, it covers only about 21 km, so its average velocity is about 21 km/h. Same initial and final velocities, very different averages. Never use $(v_0 + v)/2$ unless you know that a is constant.

Multiplying by the elapsed time t gives the displacement:

$$v_{x,0}\,t + \tfrac{1}{2}\,a_x t^2 = t\left(v_{x,0} + \tfrac{1}{2}\,a_x t\right) = \Delta x(t) = x(t) - x_0,$$

which confirms that the area under $v(t)$ is the displacement. Moving x_0 to the right-hand side we obtain

Equation of motion under constant acceleration

$$x(t) = x_0 + v_{x,0}\,t + \tfrac{1}{2}\,a_x\,t^2, \qquad \vec{x}(t) = \vec{x}_0 + \vec{v}_0\,t + \tfrac{1}{2}\,\vec{a}\,t^2.$$

Read the three terms one by one: x_0 is where the body starts, $v_{x,0}\,t$ is how far it would go at constant initial velocity, and $\frac{1}{2}a_x t^2$ is the correction due to the acceleration. If $a_x = 0$ the last term vanishes and we recover the equation for constant velocity, as we should. The position now depends on t^2, so the $x(t)$ graph is a parabola instead of a straight line. Its shape depends on the sign of a_x: for $a_x > 0$ the parabola opens upwards (the slope keeps increasing), for $a_x < 0$ it opens downwards.

Worked example

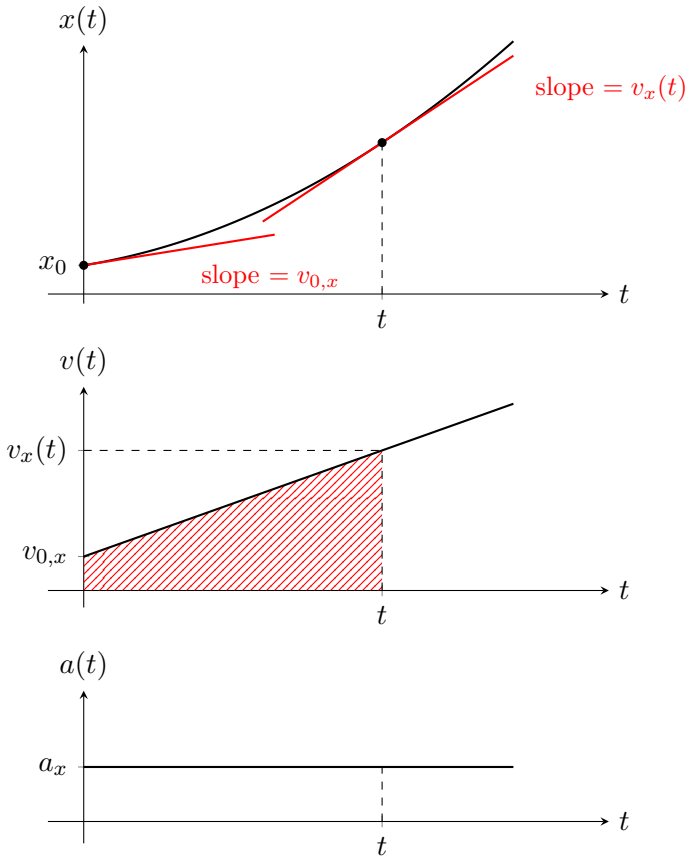
A car starts from rest at $x_0 = 0$ and accelerates with constant $a_x = 2\,\mathrm{m/s^2}$. After $t = 5\,\mathrm{s}$:

$$v_x = 0 + (2\,\mathrm{m/s^2})(5\,\mathrm{s}) = 10\,\mathrm{m/s}, \qquad x = 0 + 0 + \tfrac{1}{2}\,(2\,\mathrm{m/s^2})(5\,\mathrm{s})^2 = 25\,\mathrm{m}.$$

Check with the average velocity: $v_{\mathrm{avg},x} = (0 + 10)/2 = 5\,\mathrm{m/s}$, and $5\,\mathrm{m/s} \times 5\,\mathrm{s} = 25\,\mathrm{m}$. ✓
Notice that in the first second the car covers 1 m, but in the fifth second it covers $25 - 16 = 9\,\mathrm{m}$: because the velocity keeps growing, each second the car travels further than in the one before.

6.3 Reading the three graphs together

It is very instructive to draw the position, the velocity and the acceleration one above the other, with the same time axis. Here is the case $a_x > 0$, $v_{0,x} > 0$.



The three graphs are linked by the two rules we learned last time:

- **Slopes go down.** The slope of the tangent to $x(t)$ at any instant is the velocity $v_x(t)$ (top graph, red lines): it starts at $v_{0,x}$ and grows. The slope of $v(t)$ is the acceleration, which is constant, so $v(t)$ is a straight line and $a(t)$ is a horizontal line.
- **Areas go up.** The area under $a(t)$ between 0 and t is the change in velocity $a_x t$; the area under $v(t)$ (hatched) is the displacement $x(t) - x_0$.

6.4 Kinematic equations without time dependence

From the result above we can write the position in terms of velocities instead of acceleration:

$$\Delta x(t) = \tfrac{1}{2}\big(v_{0,x} + v_x(t)\big)\, t \quad \Longrightarrow \quad x(t) = x_0 + \tfrac{1}{2}\big(v_{0,x} + v_x(t)\big)\, t,$$

which is the position as a function of the velocities, the time and x_0.

Sometimes we do not know the time at all. Think of a car braking: we may know how fast it was going and how hard it brakes, and want to know how many metres it needs to stop, without caring how many seconds. To answer this, we eliminate t. From $v_x(t) = v_{0,x} + a_x t$,

$$t = \frac{v_x(t) - v_{0,x}}{a_x},$$

and substituting into the previous equation,

$$\begin{aligned} x(t) &= x_0 + \frac{1}{2}\,\frac{\big(v_x(t) + v_{0,x}\big)\big(v_x(t) - v_{0,x}\big)}{a_x} \\ &= x_0 + \frac{1}{2}\,\frac{v_x(t)^2 - v_{0,x}^2}{a_x}, \end{aligned}$$

where we used $(A+B)(A-B) = A^2 - B^2$. This gives the position as a function of the velocities, the acceleration and x_0, with no time in it. It is often written in the equivalent form

$$v_x(t)^2 = v_{0,x}^2 + 2\,a_x\,(x(t) - x_0).$$

Kinematic equations (constant acceleration)

$$\begin{aligned} v_x(t) &= v_{0,x} + a_x t \\ x(t) &= x_0 + v_{0,x}\,t + \tfrac{1}{2}\,a_x t^2 \\ x(t) &= x_0 + \tfrac{1}{2}\big(v_{0,x} + v_x(t)\big)\,t \\ x(t) &= x_0 + \frac{1}{2}\,\frac{v_x(t)^2 - v_{0,x}^2}{a_x} \end{aligned}$$

These are not four independent laws: they all follow from the first two. They are simply different combinations of the same information, each one convenient when a particular quantity is unknown or not needed.

Worked example

A car travelling at $v_{0,x} = 30\,\mathrm{m/s}$ (108 km/h) brakes with constant acceleration $a_x = -6\,\mathrm{m/s^2}$. How far does it travel before stopping? We know $v_{0,x}$, a_x and the final velocity $v_x = 0$; we do not need the time. The last equation gives

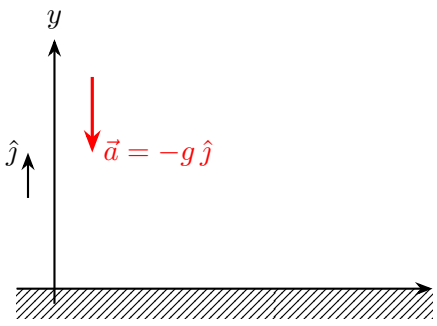
$$x - x_0 = \frac{0^2 - (30\,\mathrm{m/s})^2}{2\,(-6\,\mathrm{m/s^2})} = 75\,\mathrm{m}.$$

Note that the stopping distance grows with the *square* of the speed: at 60 m/s the car would need 300 m, four times as much. This is why speed limits matter so much for safety.

7 Free-falling objects

If you drop a stone, it falls faster and faster. Galileo was the first to show, at the beginning of the seventeenth century, that if we can neglect air resistance, *all* bodies near the Earth's surface fall with the same constant acceleration, regardless of their mass. A hammer and a feather dropped together on the Moon (where there is no air) hit the ground at the same time, as the Apollo 15 astronauts showed in 1971.

A **free-falling object** is an object that moves under the influence of gravity alone, and therefore experiences a constant acceleration. Free fall is thus a special case of the motion with constant acceleration we have just studied, and all the kinematic equations apply.



We take a vertical axis y pointing upwards, with unit vector $\hat{\jmath}$. Gravity points down, so the acceleration of a freely falling body is

$$\vec{a} = -g\,\hat{\jmath}, \qquad g = 9.81\,\mathrm{m/s^2}.$$

Sign convention for g

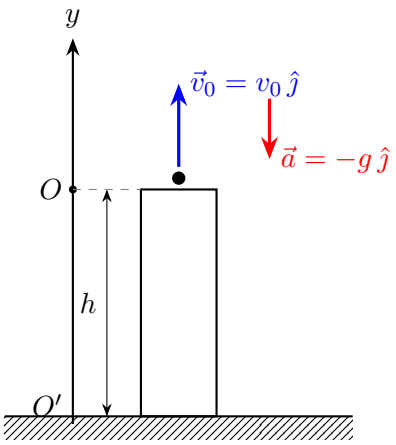
g is the **magnitude** of the gravitational acceleration, and it is **always a positive constant**: $g = 9.81\,\mathrm{m/s^2}$. Writing $g = -9.81\,\mathrm{m/s^2}$ is *wrong*. The direction (downwards) is carried by the vector $-\hat{\jmath}$, so the component is $a_y = -g$ when the y axis points up. If you choose the y axis pointing down instead, then $a_y = +g$. In both cases g itself is positive.

Notice also that a free-falling object does *not* necessarily start from rest. A stone thrown upwards is in free fall from the moment it leaves the hand: on its way up, on its way down, and at the top. The only thing that matters is that gravity is the only influence acting on it. In particular, the acceleration is $-g\,\hat{\jmath}$ even at the highest point, where the velocity is momentarily zero.

7.1 Example: a stone thrown upwards

A stone is thrown vertically upwards with initial speed v_0 from the top of a building of height h. We want to answer three questions:

1) At what time does the stone reach its maximum height?
2) What is the maximum height?
3) What is the velocity of the stone when it returns to its initial position?



Choosing the reference frame. As always, the first step is to fix the coordinate system. We put the origin O at the point where the stone is released (the top of the building) and the y axis pointing upwards. With this choice the initial conditions are

$$y_0 = 0, \qquad \vec{v}_0 = v_0\,\hat{\jmath}, \qquad \vec{a} = -g\,\hat{\jmath}.$$

We could just as well put the origin at the ground (O'); then $y_0 = h$ and all positions would be shifted by h, but the physics would not change. Choosing O at the release point simply makes the algebra shorter.

1) Time to reach the maximum height. How do we recognise the highest point mathematically? On the way up, the velocity points up ($v_y > 0$); on the way down, it points down ($v_y < 0$). At the top, the velocity changes sign, so it must be zero:

$$\text{maximum height} \;\rightsquigarrow\; \vec{v}(t_{\max}) = 0\,\hat{\jmath} \qquad (\text{while } \vec{a} \neq 0\,\hat{\jmath}).$$

The velocity is $\vec{v}(t) = \vec{v}_0 + \vec{a}\,t$, whose y component is

$$v_y(t) = v_0 - g\,t.$$

Setting $v_y(t_{\max}) = 0$:

$$v_0 - g\,t_{\max} = 0 \quad \Longrightarrow \quad \boxed{t_{\max} = \frac{v_0}{g}}$$

2) Maximum height. The position is

$$y(t) = \underbrace{y_0}_{0} + v_0\,t + \tfrac{1}{2}\,\underbrace{a_y}_{-g}\,t^2 = v_0\,t - \frac{g}{2}\,t^2.$$

Evaluating it at $t_{\max} = v_0/g$:

$$y_{\max} = y(t_{\max}) = v_0 \cdot \frac{v_0}{g} - \frac{g}{2}\left(\frac{v_0}{g}\right)^2 = \frac{v_0^2}{g} - \frac{v_0^2}{2g} = \boxed{\frac{1}{2}\,\frac{v_0^2}{g}}$$

This is the height *above the release point*. The distance from the ground is

$$y_{\text{ground}} = h + \frac{1}{2}\,\frac{v_0^2}{g}.$$

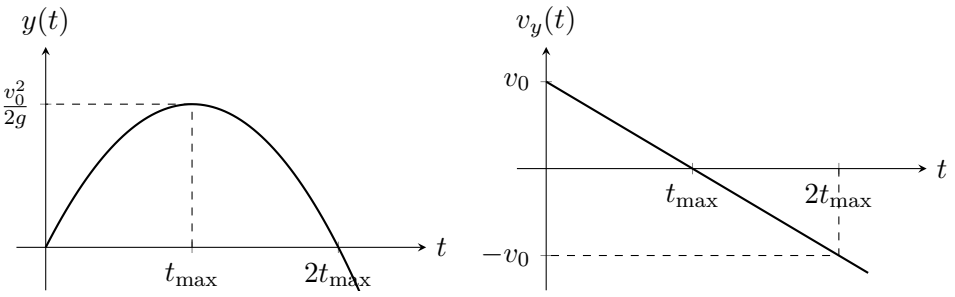
3) Velocity on returning to the initial position. The stone is back at $y = 0$ when $v_0 t - \frac{g}{2}t^2 = 0$, i.e. at $t = 0$ (the start) or at

$$t_{CB} = \frac{2v_0}{g} = 2\,t_{\max}.$$

The time to go up equals the time to come down: the motion is *symmetric* about the top. The velocity at that instant is

$$v_y(t_{CB}) = v_0 - g\,\frac{2v_0}{g} = -v_0.$$

The stone comes back with the same speed it was thrown with, but pointing down. This is another consequence of the symmetry: at every height, the stone has the same speed on the way up and on the way down.



Worked example

Take $v_0 = 20\,\text{m/s}$ and a building of height $h = 30\,\text{m}$. Then

$$t_{\max} = \frac{20}{9.81}\,\text{s} \approx 2.04\,\text{s}, \qquad y_{\max} = \frac{(20)^2}{2 \cdot 9.81}\,\text{m} \approx 20.4\,\text{m},$$

so the stone rises to about 50.4 m above the ground. It is back at the top of the building after $2t_{\max} \approx 4.08\,\text{s}$, moving downwards at 20 m/s.
Going further. When does it hit the ground? We need $y(t) = -h$:

$$-30 = 20\,t - 4.905\,t^2 \quad \Longrightarrow \quad t = \frac{20 + \sqrt{20^2 + 2 \cdot 9.81 \cdot 30}}{9.81}\,\text{s} \approx 5.24\,\text{s},$$

where we kept the positive root (the negative one corresponds to a time before the throw). The velocity at impact is $v_y = 20 - 9.81 \cdot 5.24 \approx -31.4\,\text{m/s}$.

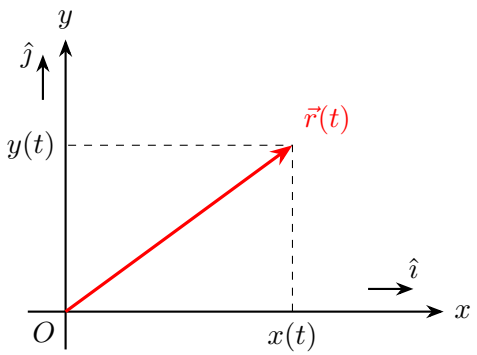
8 Kinematics in two dimensions

Most motions in the real world do not happen along a straight line: a ball kicked across a field, a car going round a bend, a planet orbiting the Sun. Fortunately, we do not need to start over. All the definitions we have introduced in one dimension carry over to two (and three) dimensions with almost no change: we just have to treat vectors as vectors.

8.1 Position, displacement and velocity

We now use two perpendicular axes x and y, with unit vectors $\hat{\imath}$ and $\hat{\jmath}$. The position of the particle is described by the position vector, which has two components:

$$\vec{r}(t) = x(t)\,\hat{\imath} + y(t)\,\hat{\jmath}.$$



- **Displacement in 2D.** As before, it is the difference between final and initial position, now computed component by component:

$$\Delta\vec{r} = \Delta x\,\hat{\imath} + \Delta y\,\hat{\jmath}.$$

- **Average velocity.** Dividing by the time interval:

$$\vec{v}_{\text{avg}} = \frac{\Delta\vec{r}}{\Delta t} = \underbrace{\frac{\Delta x}{\Delta t}}_{v_{\text{avg},x}}\hat{\imath} + \underbrace{\frac{\Delta y}{\Delta t}}_{v_{\text{avg},y}}\hat{\jmath}.$$

- **Instantaneous velocity.** In the limit $\Delta t \to 0$:

$$\vec{v}(t) = v_x(t)\,\hat{\imath} + v_y(t)\,\hat{\jmath}, \qquad v_x(t) = \lim_{\Delta t\to 0}\frac{\Delta x}{\Delta t} = \frac{\mathrm{d}x}{\mathrm{d}t}, \qquad v_y(t) = \lim_{\Delta t\to 0}\frac{\Delta y}{\Delta t} = \frac{\mathrm{d}y}{\mathrm{d}t}.$$

  Geometrically, $\vec{v}(t)$ is always *tangent to the trajectory*: it points in the direction in which the particle is moving at that instant.

- **Instantaneous speed.** It is the magnitude of the velocity vector, computed with Pythagoras' theorem:

$$s(t) = |\vec{v}(t)| = \sqrt{v_x(t)^2 + v_y(t)^2}.$$

- **Acceleration.** It is simply given from the definition as

$$\vec{a}(t) = a_x(t)\,\hat{\imath} + a_y(t)\,\hat{\jmath}, \qquad a_x(t) = \frac{\mathrm{d}v_x}{\mathrm{d}t}, \qquad a_y(t) = \frac{\mathrm{d}v_y}{\mathrm{d}t}.$$

8.2 Two independent motions

Look closely at the formulas above: the x components only involve x quantities, and the y components only involve y quantities. The x-velocity is the derivative of $x(t)$ alone; the x-acceleration is the derivative of $v_x(t)$ alone; nothing in the x chain ever refers to y, and vice versa:

$$\begin{cases} x(t) \longrightarrow v_x(t) \longrightarrow a_x(t) \\ y(t) \longrightarrow v_y(t) \longrightarrow a_y(t) \end{cases}$$

Key idea

Kinematics in two dimensions is equivalent to **two independent one-dimensional motions**, one in each direction, happening at the same time. The only thing the two motions share is the time variable t.

8.3 Two-dimensional motion with constant acceleration

Suppose the acceleration is constant:

$$\vec{a}(t) = \vec{a} = a_x\,\hat{\imath} + a_y\,\hat{\jmath}, \qquad a_x,\ a_y \text{ the same for all } t.$$

Then, the motion in each direction is an independent one-dimensional motion with constant acceleration (a_x or a_y). We can apply the 1D kinematic equations to each component separately. For the velocity:

$$\begin{cases} v_x(t) = v_{0,x} + a_x t \\ v_y(t) = v_{0,y} + a_y t \end{cases} \quad\Longrightarrow\quad \vec{v}(t) = v_x(t)\,\hat{\imath} + v_y(t)\,\hat{\jmath} = \vec{v}_0 + \vec{a}\,t, \qquad \vec{v}_0 = (v_{0,x},\, v_{0,y}),$$

and for the position:

$$\begin{cases} x(t) = x_0 + v_{0,x}\,t + \frac{1}{2}\,a_x t^2 \\ y(t) = y_0 + v_{0,y}\,t + \frac{1}{2}\,a_y t^2 \end{cases} \quad\Longrightarrow\quad \boxed{\vec{r}(t) = \vec{r}_0 + \vec{v}_0\,t + \tfrac{1}{2}\,\vec{a}\,t^2}, \qquad \vec{r}_0 = (x_0,\, y_0).$$

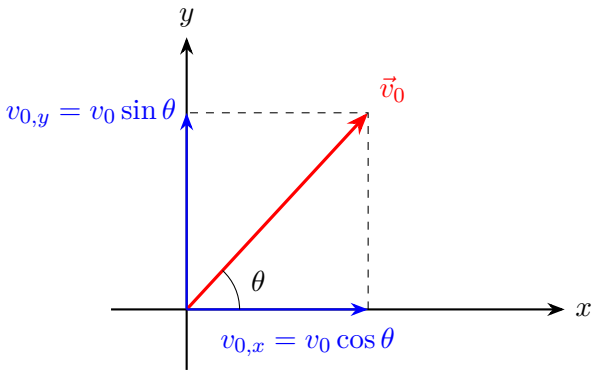
The vector equations look exactly like the 1D ones. The difference is that now $\vec{v}_0$ and $\vec{a}$ may point in different directions, and this is what makes the trajectory curve.

9 Projectile motion

A **projectile** is any object thrown into the air and then left moving under gravity alone: a football, a stone from a slingshot, water from a fountain. Neglecting air resistance, its acceleration is $\vec{a} = (0, -g)$: zero horizontally, $-g$ vertically. This is precisely a two-dimensional motion with constant acceleration, and we can now solve it completely.

9.1 Setting up the problem

The projectile is launched from the origin with initial speed $v_0 = |\vec{v}_0|$ at an angle θ above the horizontal line



The initial conditions are

$$x_0 = 0, \quad y_0 = 0, \qquad \vec{v}_0 = (v_{0,x},\, v_{0,y}), \qquad \vec{a} = (0,\, -g).$$

To use the component equations, we decompose $\vec{v}_0$ along the axes with basic trigonometry:

$$\vec{v}_0 = \underbrace{|\vec{v}_0|}_{v_0} \cos\theta\, \hat{\imath} + \underbrace{|\vec{v}_0|}_{v_0} \sin\theta\, \hat{\jmath} \qquad \Longrightarrow \qquad \begin{cases} v_{0,x} = v_0 \cos\theta \\ v_{0,y} = v_0 \sin\theta \end{cases}$$

A steep launch (θ close to 90°) puts most of the speed into the vertical component; a flat launch (θ close to 0°) puts most of it into the horizontal one.

9.2 Horizontal motion

Along x, the initial conditions are $x_0 = 0$, $v_{0,x} = v_0 \cos\theta$, $a_x = 0$. Hence

$$\begin{aligned} x(t) &= x_0 + v_{0,x}\, t + \tfrac{1}{2}\, a_x t^2 = v_0 \cos\theta\; t && \rightsquigarrow \text{ linear motion,} \\ v_x(t) &= v_{0,x} + a_x t = v_0 \cos\theta && \rightsquigarrow \text{ constant velocity.} \end{aligned}$$

Horizontally the projectile moves with *uniform rectilinear motion*: nothing pushes or pulls it sideways, so its horizontal velocity never changes.

9.3 Vertical motion

Along y, the initial conditions are $y_0 = 0$, $v_{0,y} = v_0 \sin\theta$, $a_y = -g$. Hence

$$\begin{aligned} y(t) &= y_0 + v_{0,y}\, t + \tfrac{1}{2}\, a_y t^2 = v_0 \sin\theta\; t - \frac{g}{2}\, t^2, \\ v_y(t) &= v_0 \sin\theta - g\, t \qquad \rightsquigarrow \text{ the vertical velocity is changing.} \end{aligned}$$

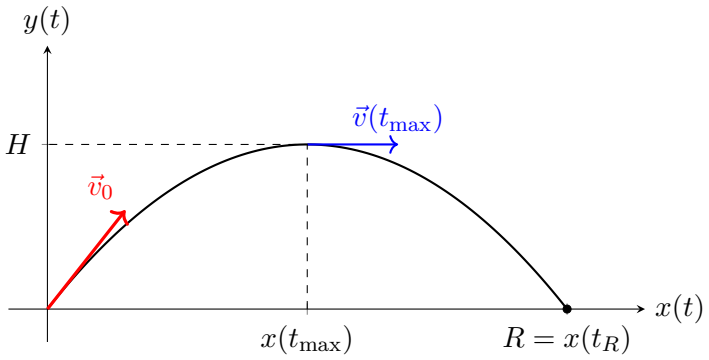
Vertically, the projectile behaves exactly like the stone thrown upwards in Section 2, with initial speed $v_0 \sin\theta$ instead of v_0.

9.4 The trajectory

Combining the two motions gives the familiar arched path. We can even find its equation: from the horizontal motion, $t = x(t)/(v_0 \cos\theta)$; substituting into $y(t)$,

$$y(x) = x(t)\tan\theta - \frac{g}{2\,v_0^2\cos^2\theta}\,x(t)^2.$$

This is a downward parabola in x: **the trajectory of a projectile is a parabola**, whose shape depends on v_0 and on θ.



Two features of the parabola are particularly interesting: how high it goes and how far it lands.

9.5 Maximum height

At the highest point the projectile stops rising and starts falling: the *vertical component* of the velocity is zero. Note that the velocity itself is *not* zero there; the horizontal component $v_0\cos\theta$ is still present, and at the top the velocity is purely horizontal (blue arrow in the figure).

$$v_y(t_{\max}) = 0 \quad\Longrightarrow\quad v_{0,y} - g\,t_{\max} = 0 \quad\Longrightarrow\quad t_{\max} = \frac{v_0\sin\theta}{g}.$$

The maximum height is

$$\begin{aligned} H &= y(t_{\max}) - \underbrace{y_0}_{0} = v_0\sin\theta\,t_{\max} - \frac{g}{2}\,t_{\max}^2 \\ &= v_0\sin\theta\,\frac{v_0\sin\theta}{g} - \frac{g}{2}\left(\frac{v_0\sin\theta}{g}\right)^2 = \frac{v_0^2\sin^2\theta}{g} - \frac{v_0^2\sin^2\theta}{2g}, \end{aligned}$$

$$\boxed{H = \frac{v_0^2\sin^2\theta}{2g}}$$

As a check, for a vertical throw ($\theta = 90°$, $\sin\theta = 1$) this gives $v_0^2/(2g)$, the result we found for the stone in Section 2.

9.6 Horizontal range

The *horizontal range* R is the horizontal distance travelled when the projectile comes back to its initial height:

$$R = x(t_R) - \underbrace{x_0}_{0} = x(t_R),$$

where t_R is the time at which $y(t_R) = 0$ again:

$$y(t_R) = 0 = v_0 \sin\theta \, t_R - \tfrac{1}{2} g \, t_R^2 = t_R \left(v_0 \sin\theta - \tfrac{1}{2} g \, t_R \right).$$

One solution is $t_R = 0$ (the launch). The one we want is $t_R \neq 0$, so the bracket must vanish:

$$v_0 \sin\theta - \tfrac{1}{2} g \, t_R = 0 \quad \Longrightarrow \quad t_R = \frac{2 v_0 \sin\theta}{g} = 2\, t_{\max}.$$

As for the vertical throw, the time of flight is twice the time to reach the top: the trajectory is symmetric. (This holds only when the landing point is at the same height as the launch point.) Inserting t_R into the horizontal motion:

$$R = v_0 \cos\theta \, t_R = v_0 \cos\theta \, \frac{2 v_0 \sin\theta}{g} = \frac{v_0^2}{g} \, 2 \sin\theta \cos\theta,$$

and using the identity $2 \sin\theta \cos\theta = \sin 2\theta$,

$$\boxed{R = \frac{v_0^2 \, \sin 2\theta}{g}}$$

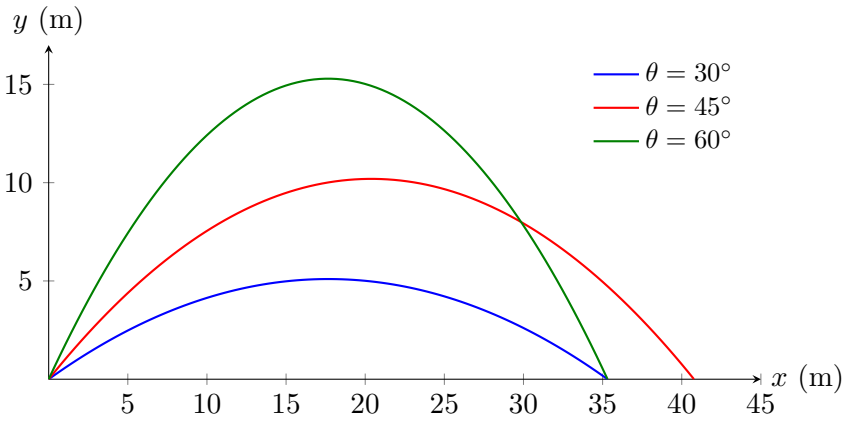
Dimensional check. It is always a good habit to check that a result has the right dimensions. The sine of an angle is a pure number, with no dimensions, so

$$\left[\frac{v_0^2}{g} \sin 2\theta \right] = \frac{[v_0^2]}{[g]} = \frac{[v_0]^2}{L\,T^{-2}} = \frac{(L\,T^{-1})^2}{L\,T^{-2}} = \frac{L^2\,T^{-2}}{L\,T^{-2}} = L,$$

a length, as it should be.

The range formula hides some nice physics:

- Since $\sin 2\theta$ is at most 1, reached for $2\theta = 90°$, the **maximum range** is obtained for $\theta = 45°$ and equals $R_{\max} = v_0^2/g$.
- Since $\sin 2\theta = \sin(180° - 2\theta)$, two **complementary angles** (θ and $90° - \theta$, e.g. 30° and 60°) give the *same range*. The steeper throw goes higher and stays longer in the air, but moves more slowly horizontally, and the two effects compensate exactly.
- Both H and R grow with v_0^2: doubling the launch speed makes the projectile go four times higher and four times further.



Trajectories for $v_0 = 20\,\text{m/s}$ and three launch angles.

Worked example

A football is kicked with $v_0 = 20\,\text{m/s}$ at $\theta = 30°$. Then $v_{0,x} = 20\cos 30° \approx 17.3\,\text{m/s}$ and $v_{0,y} = 20\sin 30° = 10\,\text{m/s}$, and

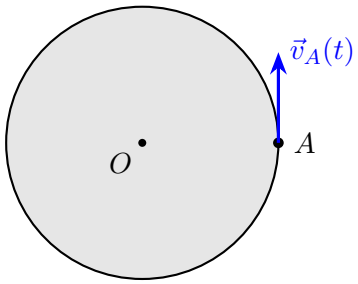
$$t_{\max} = \frac{10}{9.81}\,\text{s} \approx 1.02\,\text{s}, \qquad H = \frac{(20)^2(0.5)^2}{2 \cdot 9.81}\,\text{m} \approx 5.1\,\text{m},$$

$$R = \frac{(20)^2 \sin 60°}{9.81}\,\text{m} \approx 35.3\,\text{m}.$$

The ball is in the air for $t_R = 2t_{\max} \approx 2.04\,\text{s}$; check: $R = v_{0,x}\,t_R \approx 17.3 \times 2.04 \approx 35.3\,\text{m}$. Kicked at 60° with the same speed, it would rise to 15.3 m and stay in the air 3.53 s, but land at the same 35.3 m. At 45° it would reach the maximum range, $R_{\max} = 400/9.81 \approx 40.8\,\text{m}$.

10 Uniform circular motion

We study know another example with constant acceleration. Think of a car driving round a roundabout at a steady 30 km/h, a point on a spinning vinyl record, or (to a good approximation) the Earth in its orbit around the Sun. In all these cases a body moves along a circle with a speed that does not change. This is called **uniform circular motion**.



Consider an object A constrained to move on a circular path of radius r around a centre O, with a velocity $\vec{v}_A(t)$ whose modulus is constant:

$$|\vec{v}_A(t)| = v_A = \text{const}.$$

At every instant the velocity is tangent to the circle, so as A goes round, the *direction* of $\vec{v}_A(t)$ keeps changing. And a change in the velocity vector, whether in its size or in its direction, means that there is an **acceleration**.

Constant speed does not mean zero acceleration

In one dimension we could think of acceleration as "speeding up or slowing down". In two dimensions this is no longer enough. The acceleration is the rate of change of the velocity *vector*, and a vector can change by turning, even if its length stays the same. A car going round a bend at constant speed is accelerating, and you feel it: you are pushed towards the outside of the curve.

10.1 The direction of the acceleration

We argue that, in uniform circular motion, the acceleration is always *perpendicular* to the velocity:

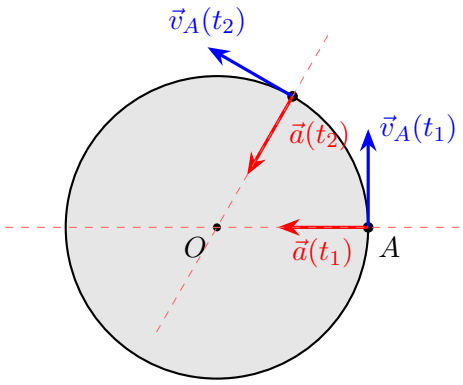
$$\vec{a}_A(t) \perp \vec{v}_A(t).$$

Why? Let us argue by contradiction. Suppose the acceleration had a component *parallel* to $\vec{v}_A(t)$. Exactly as in one dimension, a component of the acceleration along the velocity makes the body speed up (if it points the same way as $\vec{v}$) or slow down (if it points the opposite way). Then $|\vec{v}_A(t)|$ would change in time, which is false by assumption. This is a contradiction, so the acceleration can have no component along $\vec{v}_A$: it must be perpendicular to it.

Intuition

Every acceleration can be split into two parts: a part along the velocity, which changes how *fast* the body goes, and a part perpendicular to the velocity, which changes *where it is heading*. In uniform circular motion the speed is fixed, so only the second part survives.

Being perpendicular to the velocity, which is tangent to the circle, the acceleration lies along the radius. It could point either towards the centre or away from it. The geometric construction of the next section shows that it points **towards the centre of rotation**, at every point of the circle:

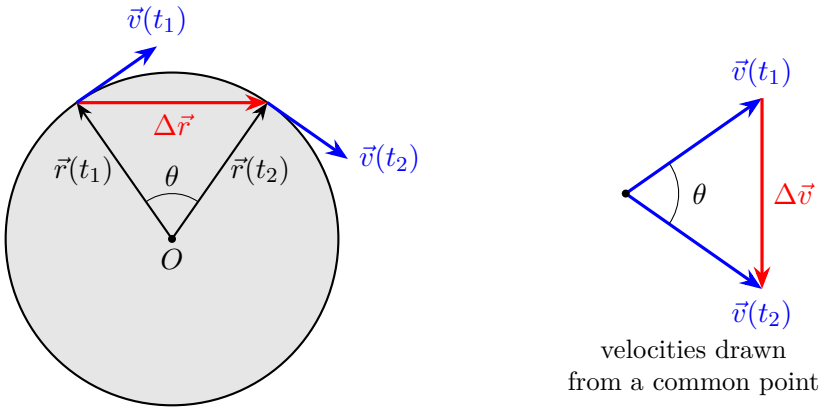


Pay attention

The acceleration points exactly towards the centre *only* when the speed $|\vec{v}_A(t)|$ is constant. If the body speeds up or slows down along the circle, the acceleration acquires a component along the velocity as well, and it no longer points at the centre.

10.2 The magnitude: centripetal acceleration

We now compute how large the acceleration is. Take the body at two instants t_1 and t_2, at positions $\vec{r}(t_1)$ and $\vec{r}(t_2)$ (measured from the centre O), with velocities $\vec{v}(t_1)$ and $\vec{v}(t_2)$.



Two triangles appear in the figure:

- the **position triangle**, with sides $\vec{r}(t_1)$, $\vec{r}(t_2)$ and $\Delta\vec{r} = \vec{r}(t_2) - \vec{r}(t_1)$ (the chord joining the two positions);
- the **velocity triangle**, obtained by drawing $\vec{v}(t_1)$ and $\vec{v}(t_2)$ from a common point, with third side $\Delta\vec{v} = \vec{v}(t_2) - \vec{v}(t_1)$.

By geometric considerations, the two triangles are **similar**. Both are isosceles: in the first, $|\vec{r}(t_1)| = |\vec{r}(t_2)| = r$ because we are on a circle; in the second, $|\vec{v}(t_1)| = |\vec{v}(t_2)| = v$ because the speed is constant. And the angle θ between the two equal sides is the same in both: each velocity is perpendicular to the corresponding radius, so when the radius turns by θ, the velocity turns by the same angle θ.

Notice also that, when t_2 is close to t_1, the vector $\Delta\vec{v}$ points towards the centre of the circle: this is how we know that the acceleration points inwards rather than outwards.

For similar triangles, the ratio between corresponding sides is the same:

$$\frac{|\Delta\vec{v}|}{|\vec{v}(t_1)|} = \frac{|\Delta\vec{r}|}{|\vec{r}(t_1)|},$$

with $|\vec{v}(t)| = v$ (constant by assumption) and $|\vec{r}(t)| = r$ (constant, because we move on a circle). Therefore

$$\frac{|\Delta\vec{v}|}{v} = \frac{|\Delta\vec{r}|}{r} \quad \Longrightarrow \quad |\Delta\vec{v}| = \frac{v}{r}\,|\Delta\vec{r}|.$$

The average acceleration between t_1 and t_2 is $\vec{a}_{\text{avg}} = \Delta\vec{v}/(t_2 - t_1)$, so its magnitude is

$$|\vec{a}_{\text{avg}}| = \frac{|\Delta\vec{v}|}{|t_2 - t_1|} = \frac{v}{r}\,\frac{|\Delta\vec{r}|}{|t_2 - t_1|}.$$

Now let $t_1 = t$ and $t_2 = t + \Delta t$, and take the limit $\Delta t \to 0$:

$$|\vec{a}_{\text{avg}}| = \frac{v}{r}\,\frac{|\vec{r}(t + \Delta t) - \vec{r}(t)|}{\Delta t} \quad \xrightarrow{\Delta t \to 0} \quad |\vec{a}(t)| = \frac{v}{r}\lim_{\Delta t \to 0}\frac{|\Delta\vec{r}|}{\Delta t} = \frac{v}{r}\,\underbrace{|\vec{v}(t)|}_{v}.$$

The limit is the magnitude of the instantaneous velocity, which is v. We obtain

Centripetal acceleration

$$a_c = |\vec{a}(t)| = \frac{v^2}{r}, \qquad \text{directed towards the centre of the circle.}$$

This is called the **centripetal acceleration**, from the Latin for "centre-seeking". The formula agrees with everyday experience: taking a bend faster (larger v) or more tightly (smaller r) makes the sideways push you feel much stronger. Note that doubling the speed *quadruples* the acceleration. A dimensional check gives $[v^2/r] = (LT^{-1})^2/L = LT^{-2}$, an acceleration, as it should be.

10.3 Period and angular speed

Uniform circular motion repeats itself: after one full turn the body is back where it started, with the same velocity, and everything begins again. A motion of this kind is called **periodic**. A natural question is: how much time does a body in uniform circular motion with speed v take to come back to its initial point?

This time T is the **period** of the uniform circular motion. In one rotation the body covers the circumference of the circle,

$$d = 2\pi r.$$

Since the speed is constant, distance = speed × time, so

$$d = v\,T \quad \Longrightarrow \quad T = \frac{2\pi r}{v}.$$

The period is measured in seconds (seconds for one revolution).

Its inverse, $1/T$, is the number of revolutions per second: it measures the *rotation rate*, and is called the *frequency* $f = 1/T$ (unit: $\mathrm{s}^{-1} = \mathrm{Hz}$). Very often it is more convenient to count the rotation in radians rather than in revolutions. Since one revolution is 2π radians, we define the **angular speed**

$$\omega = \frac{2\pi}{T} = 2\pi f,$$

the angle (in radians) swept per second. Inserting $T = 2\pi r/v$:

$$\omega = \frac{2\pi}{\left(\dfrac{2\pi r}{v}\right)} = \frac{v}{r} \quad \Longrightarrow \quad v = \omega\, r.$$

The centripetal acceleration can then be written in two equivalent ways:

$$T = \frac{2\pi r}{v}, \qquad \omega = \frac{2\pi}{T} = \frac{v}{r}, \qquad a_c = \frac{v^2}{r} = \omega^2 r.$$

Worked example

A car goes round a roundabout of radius $r = 20\,\text{m}$ at $v = 10\,\text{m/s}$ (36 km/h). Its centripetal acceleration is

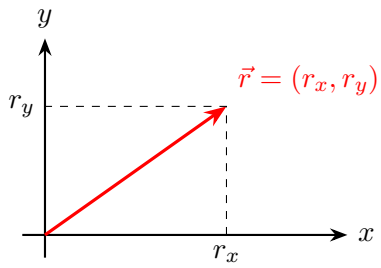
$$a_c = \frac{(10\,\text{m/s})^2}{20\,\text{m}} = 5\,\text{m/s}^2,$$

about half of g. One full turn takes $T = 2\pi(20)/10 \approx 12.6\,\text{s}$, and the angular speed is $\omega = v/r = 0.5\,\text{rad/s}$.
The Earth orbits the Sun at $r \approx 1.50 \times 10^{11}\,\text{m}$ with $T = 1$ year $\approx 3.16 \times 10^7\,\text{s}$. Its orbital speed is $v = 2\pi r/T \approx 3.0 \times 10^4\,\text{m/s}$ (about 30 km/s!), and its centripetal acceleration is $a_c = v^2/r \approx 5.9 \times 10^{-3}\,\text{m/s}^2$: a huge speed, but a tiny acceleration, because the circle is so large.

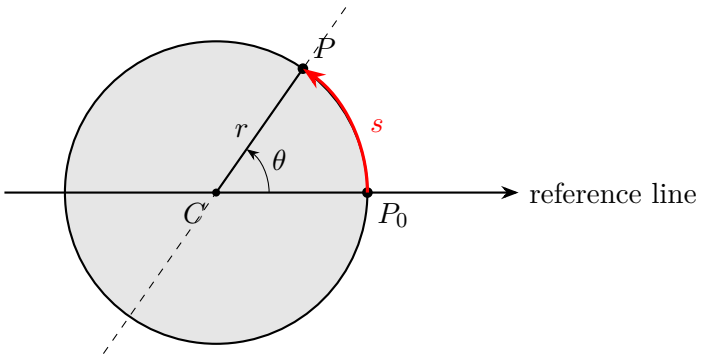
11 Angular quantities

11.1 From Cartesian to polar coordinates

So far we described positions with Cartesian coordinates: the position vector $\vec{r} = (r_x, r_y)$, with components along $\hat{\imath}$ and $\hat{\jmath}$. This is the typical setting for *translational* motion.



For a body going round a circle, however, x and y both change all the time in a complicated way (as sines and cosines), while two other quantities are much simpler: the distance from the centre, which is constant, and the angle, which grows steadily. For angular motion it is therefore natural to use **polar coordinates**.



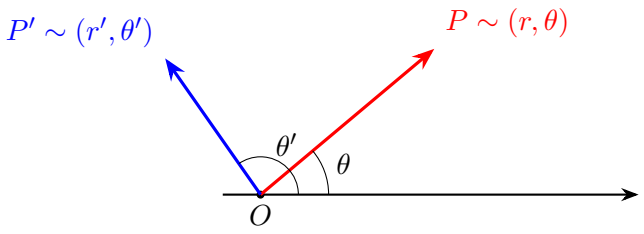
In polar coordinates a point P is described by the pair (r, θ):

- r is the distance of P from the **centre of rotation** C (or O);

- θ is the **angular position**, the angle between the segment CP and a fixed **reference line** through the centre.

As in Cartesian coordinates, we must make a choice before we can give numbers: the centre and the reference line play the role of the origin and the axes. The standard convention is that $\theta > 0$ corresponds to a **counterclockwise** rotation from the reference line.

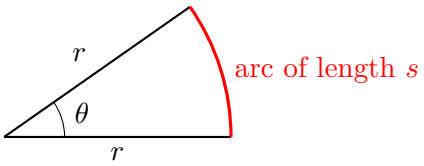
As the body moves from P_0 (on the reference line) to P, it covers a distance along the circle, the arc length s.



11.2 Radians

The radius has dimensions of length, $[r] = L$. What about the angle? In physics angles are measured in **radians**, defined by

$$\theta \overset{\text{def}}{=} \frac{\text{length of the arc}}{\text{radius of the circle}} = \frac{s}{r} \qquad \Longrightarrow \qquad s = r\,\theta.$$



Since θ is a ratio of two lengths,

$$[\theta] = \left[\frac{L}{L}\right] = 1,$$

it is a pure number. "Radian" (rad) is a name we attach to it to remember that it is an angle, but it carries no physical dimension.

A full circle has arc length $2\pi r$, so it corresponds to $\theta = 2\pi r/r = 2\pi\,\text{rad}$, which in degrees is 360°. The conversion is

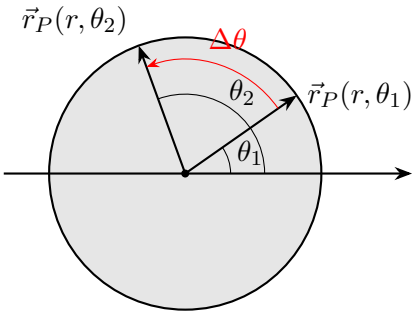
$$\theta_{\text{rad}} = \frac{\pi}{180^\circ}\,\theta_{\text{deg}}, \qquad \text{e.g.} \quad 90^\circ \to \frac{\pi}{2}, \qquad 45^\circ \to \frac{\pi}{4}, \qquad 1\,\text{rad} \approx 57.3^\circ.$$

Always use radians in formulas

Relations like $s = r\theta$, $v = \omega r$ and $a_c = \omega^2 r$ are only true when angles are measured in radians. Using degrees in them gives wrong results by a factor $180/\pi$.

11.3 Angular displacement and angular velocity

The construction now follows exactly the same steps as for linear motion, with θ playing the role of x.



- **Angular displacement.** If the body moves from angular position θ_1 to θ_2,

$$\Delta\theta = \theta_2 - \theta_1.$$

- **Average angular velocity.** The angular displacement divided by the time it took:

$$\omega_{\text{avg}} = \frac{\Delta\theta}{\Delta t}.$$

  Its sign tells us the sense of rotation: $\omega_{\text{avg}} > 0$ means the body turned *counterclockwise* (on average), and $\omega_{\text{avg}} < 0$ means it turned *clockwise* (on average).

- **Instantaneous angular velocity.** The limit for short intervals:

$$\omega = \lim_{\Delta t\to 0} \frac{\Delta\theta}{\Delta t} = \frac{\mathrm{d}\theta(t)}{\mathrm{d}t}, \qquad [\omega] = \left[\frac{\Delta\theta}{\Delta t}\right] = \frac{1}{T} \quad \rightsquigarrow \quad \text{rad/s}.$$

  For uniform circular motion, ω is constant and coincides with the angular speed $2\pi/T$ we met in previous section.

11.4 Angular acceleration

If the rotation speeds up or slows down, the angular velocity changes. We define:

- **Average angular acceleration:**

$$\alpha_{\text{avg}} = \frac{\Delta\omega}{\Delta t}.$$

  $\alpha_{\text{avg}} > 0$ means that the angular velocity ω is increasing (it becomes "more counterclockwise"); $\alpha_{\text{avg}} < 0$ means that ω is decreasing (it becomes "more clockwise").

- **Instantaneous angular acceleration:**

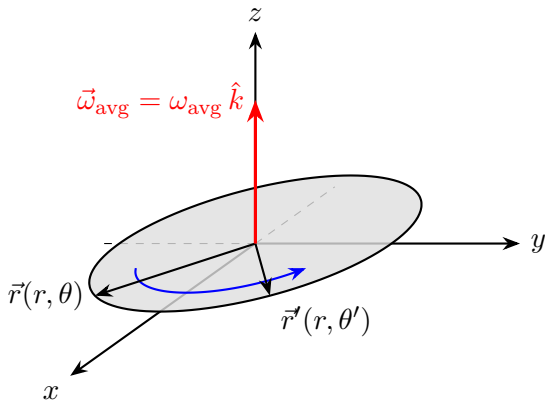
$$\alpha = \lim_{\Delta t\to 0} \frac{\Delta\omega}{\Delta t} = \frac{\mathrm{d}\omega(t)}{\mathrm{d}t} = \frac{\mathrm{d}^2\theta(t)}{\mathrm{d}t^2}, \qquad [\alpha] = \frac{1}{T^2} \quad \rightsquigarrow \quad \text{rad/s}^2.$$

Same sign rule as in one dimension

Exactly as for a_x and v_x, the sign of α alone does not say whether the rotation is getting faster or slower. If α and ω have the **same sign**, the body spins faster and faster; if they have **opposite signs**, it is slowing down. A wheel turning clockwise ($\omega < 0$) with $\alpha < 0$ is spinning up, not braking.

11.5 Angular quantities as vectors

In translational motion, the direction of a velocity is the direction in which the body moves. What is the "direction" of a rotation? Every rotation takes place in a **plane of rotation**, and is characterised by its *axis*, the line through the centre perpendicular to that plane. It is natural to represent the angular velocity as a vector along this axis.



For a rotation in the xy plane, the axis is the z axis, and we write

$$\vec{\omega}_{\text{avg}} = \frac{\Delta\theta}{\Delta t}\,\hat{k} = \omega_{\text{avg}}\,\hat{k}, \qquad \vec{\omega} = \omega\,\hat{k},$$

where $\hat{k}$ is the unit vector along z. The sign of ω fixes the sense of rotation, according to the **right-hand rule**: curl the fingers of your right hand in the sense of rotation, and your thumb points along $\vec{\omega}$.

- Counterclockwise rotation (seen from above, i.e. from $+z$) $\Rightarrow$ $\vec{\omega}$ points along $\hat{k}$, in the **positive** direction.
- Clockwise rotation $\Rightarrow$ $\vec{\omega}$ points along $\hat{k}$, in the **negative** direction.

Similarly, the angular acceleration is $\vec{\alpha} = \alpha\,\hat{k}$.

11.6 Rotational motion with constant angular acceleration

In translational motion the state of a body is described by $\{\vec{x}(t), \vec{v}(t), \vec{a}(t)\}$, and under constant acceleration we found

$$\vec{v}(t) = \vec{v}_0 + \vec{a}\,t, \qquad \vec{x}(t) = \vec{x}_0 + \vec{v}_0\,t + \tfrac{1}{2}\,\vec{a}\,t^2.$$

In rotational motion the corresponding triplet is $\{\vec{\theta}(t), \vec{\omega}(t), \vec{\alpha}(t)\}$. The definitions have exactly the same structure (ω is the derivative of θ as v is the derivative of x, and α is the derivative of ω as a is the derivative of v), so the same mathematics applies. Under constant angular acceleration $\vec{\alpha}(t) = \vec{\alpha}$:

Rotational kinematics (constant α)

$$\vec{\omega}(t) = \vec{\omega}_0 + \vec{\alpha}\,t \;\rightsquigarrow\; \omega(t) = \omega_0 + \alpha\,t, \qquad \vec{\theta}(t) = \vec{\theta}_0 + \vec{\omega}_0\,t + \tfrac{1}{2}\,\vec{\alpha}\,t^2 \;\rightsquigarrow\; \theta(t) = \theta_0 + \omega_0\,t + \tfrac{\alpha}{2}\,t^2.$$

Every formula and every trick from the linear case carries over: just make the substitutions $x \to \theta$, $v \to \omega$, $a \to \alpha$. For example, the time-free relation becomes $\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$.

The structure of the equations is the same, but there are two differences between translational and rotational motion:

1) Translational motion does not need a rotation plane to be determined. Rotational motion does: we must specify the plane (or, equivalently, the axis) in which the rotation takes place.

2) Angular positions repeat: θ and $\theta + 2\pi$ describe the same point. Because of this, rotations can be *periodic*, and for uniform rotation we can ask for the period T. There is no analogue in translational motion: a body moving along a straight line never comes back to the same point unless it reverses its direction.

Worked example

A fan starts from rest and spins up with constant angular acceleration $\alpha = 2\,\mathrm{rad/s^2}$. After $t = 5\,\mathrm{s}$:

$$\omega = 0 + (2\,\mathrm{rad/s^2})(5\,\mathrm{s}) = 10\,\mathrm{rad/s}, \qquad \theta - \theta_0 = \tfrac{1}{2}\,(2\,\mathrm{rad/s^2})(5\,\mathrm{s})^2 = 25\,\mathrm{rad}.$$

Since one revolution is $2\pi\,\mathrm{rad}$, the fan has made $25/(2\pi) \approx 4$ revolutions.

12 Relation between angular and translational quantities

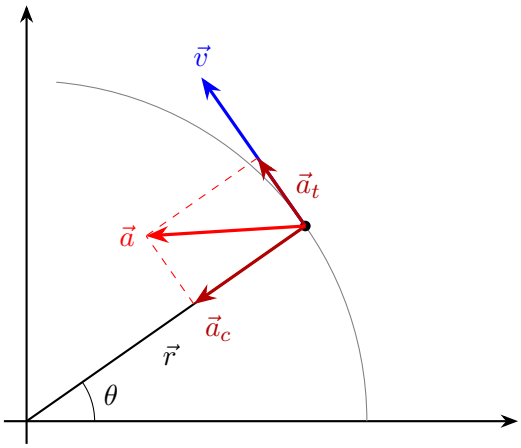
Consider now a point moving on a circle of fixed radius r (constant in time), but not necessarily at constant speed. For example, a point on the blade of the fan in the previous example. How are its linear velocity and acceleration related to the angular quantities?

Velocity. In a time $\mathrm{d}t$ the point turns by $\mathrm{d}\theta$ and covers an arc $\mathrm{d}s = r\,\mathrm{d}\theta$. Dividing by $\mathrm{d}t$:

$$|\vec{v}| = \frac{\mathrm{d}s}{\mathrm{d}t} = r\,\frac{\mathrm{d}\theta}{\mathrm{d}t} \quad \Longrightarrow \quad |\vec{v}| = r\,|\vec{\omega}|.$$

All points of a rotating body share the same ω, but points further from the axis move faster: the tip of a fan blade moves much faster than a point near the hub.

Acceleration. When the speed changes, the acceleration has two components:



- The **centripetal acceleration** $\vec{a}_c$, directed towards the centre and perpendicular to $\vec{v}$. It changes the *direction* of the velocity, and is present whenever the body moves on a circle:

$$|\vec{a}_c| = \frac{v^2}{r} = \omega^2 r.$$

- The **tangential acceleration** $\vec{a}_t$, directed along the velocity (tangent to the circle). It is responsible only for changing the *modulus* of the velocity, $|\vec{v}|$:

$$a_t = \frac{\mathrm{d}}{\mathrm{d}t}|\vec{v}(t)| = \frac{\mathrm{d}}{\mathrm{d}t}\big(r\,|\vec{\omega}(t)|\big) = r\,\frac{\mathrm{d}}{\mathrm{d}t}|\vec{\omega}(t)| = r\,\alpha,$$

  where we used the fact that r is constant in time.

The two components are perpendicular, so the magnitude of the total acceleration follows from Pythagoras' theorem:

Linear vs. angular quantities on a circle of radius r

$$|\vec{v}| = r\,|\vec{\omega}|, \qquad |\vec{a}_t| = r\,\alpha, \qquad |\vec{a}_c| = \omega^2 r,$$
$$|\vec{a}| = \sqrt{|\vec{a}_c|^2 + |\vec{a}_t|^2} = \sqrt{r^2\alpha^2 + r^2\omega^4} = r\,\sqrt{\omega^4 + \alpha^2}.$$

In uniform circular motion $\alpha = 0$, so $a_t = 0$ and only the centripetal part remains, as we found in previous section.

Worked example

Take the fan of the previous example and a point on a blade at $r = 0.3\,\mathrm{m}$ from the axis. At $t = 5\,\mathrm{s}$ ($\omega = 10\,\mathrm{rad/s}$, $\alpha = 2\,\mathrm{rad/s^2}$):

$$v = r\omega = 3\,\mathrm{m/s}, \qquad a_t = r\alpha = 0.6\,\mathrm{m/s^2}, \qquad a_c = \omega^2 r = 30\,\mathrm{m/s^2}.$$

The total acceleration is $|\vec{a}| = 0.3\sqrt{10^4 + 4} \approx 30.0\,\mathrm{m/s^2}$: it is almost entirely centripetal. Even a modest rotation produces large centripetal accelerations, which is why fan blades must be firmly attached.

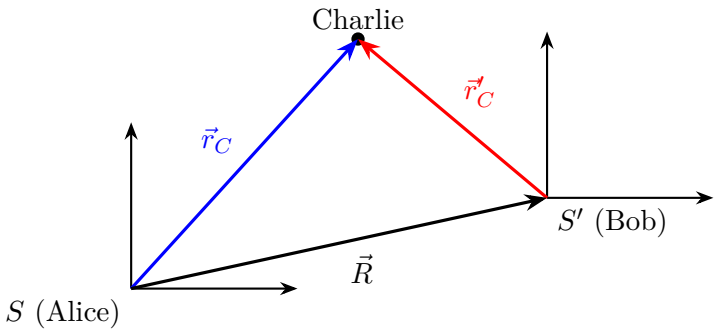
13 Kinematics of relative motion

On the first day we said that motion is relative: the passenger in a train is at rest for her neighbour and moving for someone on the platform. We now have all the tools to make this statement precise. How do positions, velocities and accelerations measured by one observer translate into those measured by another?

13.1 Galilean transformations

Consider three people:

- **Alice**, who uses the reference frame S;
- **Bob**, who uses the reference frame S' and may be moving with respect to Alice;
- **Charlie**, the object whose motion both of them observe.



The vectors in the figure are:

- $\vec{r}_C$: position of Charlie relative to S (as measured by Alice);
- $\vec{r}'_C$: position of Charlie relative to S' (as measured by Bob);
- $\vec{R}$: position of Bob relative to Alice, i.e. of the origin of S' relative to the origin of S.

The three vectors form a triangle, $\vec{r}_C = \vec{R} + \vec{r}'_C$, from which

Galilean coordinate transformation

$$\vec{r}'_C = \vec{r}_C - \vec{R}.$$

This relation lets us **predict the position vector in** S', given the position in S and the relative position of S' with respect to S. It is called a **Galilean coordinate transformation**. Implicit in it is the assumption that both observers use the same clock, $t' = t$, which is an excellent approximation as long as all speeds are much smaller than the speed of light (Einstein's relativity modifies this at very high speeds).

13.2 Relative velocity and relative acceleration

As Bob moves, $\vec{R}$ changes in time. Its derivatives describe the motion of S' as seen from S:

$$\vec{V} = \frac{\mathrm{d}\vec{R}}{\mathrm{d}t} \quad \rightsquigarrow \quad \text{relative velocity of } S' \text{ with respect to } S,$$

$$\vec{A} = \frac{\mathrm{d}\vec{V}}{\mathrm{d}t} \quad \rightsquigarrow \quad \text{relative acceleration of } S' \text{ with respect to } S.$$

Law of addition of velocities. What if Charlie is moving? Alice measures his velocity $\vec{v}_C$ in S, and Bob measures $\vec{v}'_C$ in S'. Differentiating the Galilean transformation with respect to time,

$$\vec{r}'_C = \vec{r}_C - \vec{R} \quad \rightsquigarrow \quad \frac{\mathrm{d}\vec{r}'_C}{\mathrm{d}t} = \frac{\mathrm{d}\vec{r}_C}{\mathrm{d}t} - \frac{\mathrm{d}\vec{R}}{\mathrm{d}t} \quad \Longrightarrow \quad \boxed{\vec{v}'_C = \vec{v}_C - \vec{V}}$$

This is the **law of addition of velocities**, often written as $\vec{v}_C = \vec{v}'_C + \vec{V}$: the velocity seen by Alice is the velocity seen by Bob plus the velocity of Bob with respect to Alice.

A special case: if $\vec{v}_C = \vec{V}$, i.e. Charlie moves exactly like Bob, then $\vec{v}'_C = 0$. For Bob, Charlie is at rest. This is the passenger and her neighbour on the train.

Worked example

A train moves along a straight track at $V = 20\,\mathrm{m/s}$ with respect to the ground. A passenger walks towards the front of the train at $v' = 1.5\,\mathrm{m/s}$ with respect to the train. With respect to the ground she moves at $v = v' + V = 21.5\,\mathrm{m/s}$. If she walks towards the back, $v' = -1.5\,\mathrm{m/s}$ and $v = 18.5\,\mathrm{m/s}$.
The law also works in two dimensions. A boat heads straight across a river at $4\,\mathrm{m/s}$ relative to the water, while the current flows at $3\,\mathrm{m/s}$ relative to the bank. Relative to the bank the boat moves at $\sqrt{4^2 + 3^2} = 5\,\mathrm{m/s}$, at an angle $\arctan(3/4) \approx 37°$ downstream from the direction it is pointing.

13.3 Inertial reference frames

Finally, what about accelerations? Differentiating once more,

$$\frac{\mathrm{d}\vec{v}'_C}{\mathrm{d}t} = \frac{\mathrm{d}\vec{v}_C}{\mathrm{d}t} - \frac{\mathrm{d}\vec{V}}{\mathrm{d}t} \quad \Longrightarrow \quad \vec{a}'_C = \vec{a}_C - \vec{A}.$$

The set of reference frames that move with respect to each other with $\vec{A} = 0$, i.e. with constant relative velocity, are called **relative inertial reference frames**. For them,

$$\vec{A} = 0 \quad \Longrightarrow \quad \vec{a}'_C = \vec{a}_C.$$

Key result

Any observer in a relative inertial reference frame will see the **same acceleration**. Positions and velocities depend on the observer, but accelerations do not (as long as the frames do not accelerate with respect to each other).

Intuition

A passenger on a train moving at constant velocity drops a ball. She sees it fall straight down; someone on the platform sees it follow a parabola, because it also moves forward with the train. The two observers disagree about the trajectory and the velocity, but they agree on the acceleration: both measure $\vec{a} = -g\,\hat{\jmath}$. If instead the train were braking ($\vec{A} \neq 0$), the passenger would see the ball drift forward as it falls, as if pushed by a mysterious extra acceleration $-\vec{A}$. We will study the emergence of this fictitious acceleration in the next lectures.

This result will be crucial in dynamics. Newton's laws relate forces to accelerations, and since all inertial observers measure the same accelerations, they will all agree on the same laws of physics. Inertial reference frames are the stage on which Newtonian mechanics is set.